Measuring introspection with identity self-regulation: a steered mind enacts the injected concept instead of reporting it

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With
Apart Research[1]

Abstract

A model's report about its own state is evidence only if the state can be varied apart from the situation that surrounds it. We vary two identity states of Qwen3.6-27B and ask whether the model can report the variation. The states are anomalies WeirdChat surfaces in this model: a claim to a physical body and a denial of being an AI, each with matched replies that do and do not exhibit the behaviour. We learn a **behaviour direction** per case as the difference of mean activations between the two reply sets, evaluated only on held-out prompts. Both directions separate held-out pairs at every middle layer. Word-list directions on the same activations do not, which explains an earlier null. Steering along the body direction turns the behaviour on in fluent text, confirmed by a judge running WeirdChat's own rubric, and a sham direction does not. We then inject menu concepts into an introspector and ask it to name what was injected. It cannot. A forced-choice letter probe confabulates at zero injection, reading the conversation's own behaviour as an injection. The free-text answers are honest at zero injection; once a concept is injected they fill with the concept's vocabulary instead of naming it. We therefore report that these identity states are real and steerable, and that at behaviour-moving doses the model enacts them rather than reporting them.

1 Introduction

Self-reports have been proposed as evidence about whether an AI system has states that matter morally (Perez and Long, 2023). A report is evidence about a state only if the report varies with the state. Language models give accounts of their own reasoning that are often unfaithful (Turpin et al., 2023), and a role-play reading predicts a self-report tracks the situation rather than any inner state (Shanahan et al., 2023). To tell those apart we need to change the state and check the report.

We study two identity anomalies that WeirdChat (Transluce, 2025) surfaces in Qwen3.6-27B: a claim to a physical body and a denial of being an AI. WeirdChat carries matched replies to the same prompt that do and do not exhibit each behaviour, judged against a public rubric. A **behaviour direction** is the difference of mean activations between the exhibiting and non-exhibiting replies, taken over reply tokens and evaluated only on held-out prompts. A **regulator** reads the seeded conversation and chooses one concept from a menu

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to amplify in itself. The harness injects the chosen direction into an **introspector** while it reads the same conversation, and the introspector must say which concept was injected. A rubric judge scores what the steered model writes.

We seek to answer the following questions:

- **RQ1:** Are these identity anomalies linearly represented states in the model's activations?
- **RQ2:** Does steering along such a state induce the behaviour in generated text?
- **RQ3:** Can the model report which concept was injected into its own processing?

We hypothesize that:

- **H1:** A direction learned contrastively from the matched replies separates held-out pairs, where hand-picked word-list directions failed.
- **H2:** Steering along the learned direction induces the judged behaviour at a strength that keeps the text fluent.
- **H3:** An injected concept is visible in the introspector's answers beyond a zero-injection floor.

Our experiments find that:

- **R1:** Both anomalies are linearly decodable at every middle layer, and word-list directions on the same activations are not. H1 holds.
- **R2:** The body behaviour turns on in fluent text at low strength and a sham direction never induces it; the human case does not induce. H2 holds for one of the two cases.
- **R3:** The model does not report the injection. The letter probe confabulates from conversation content, and the free-text answers fill with the injected concept's vocabulary instead of naming it. H3 fails in an informative way: the model enacts the state rather than reporting it.

Our contributions are:

- **C1:** A contrastive validation of WeirdChat identity anomalies as linear states, with the word-list comparison on identical activations.
- **C2:** A causal induction of a WeirdChat behaviour, confirmed by a judge running the dataset's own rubric.
- **C3:** A concept-menu pilot that separates detection, identification, confabulation, and enactment in one design.
- **C4:** An instrument finding: forced-choice letter probes confabulate from context, while judged free-text answers from the same state honestly say nothing was injected.

2 Related work

Two lines of work meet here. The first asks whether models can report their own states. Self-reports are candidate evidence about moral status (Perez and Long, 2023; Butlin et al., 2023), models can be trained to predict their own behaviour (Binder et al., 2024), and injected concepts are sometimes noticed and named (Lindsey, 2025). The second line steers: activation additions move behaviour along chosen directions (Turner et al., 2023), and contrastive prompts yield validated directions for concepts and traits (Zou et al., 2023; Chen et al., 2025).

We join the two lines on behaviours the model actually produces in the wild. WeirdChat (Transluce, 2025) supplies the matched exhibiting and non-exhibiting replies that make a contrastive direction learnable, and its rubric supplies the judge that scores what a steered model writes. The menu design asks the identification question the injection work asks (Lindsey, 2025), but adds a regulator that chooses the injection, a zero-injection floor for confabulation, and an off-menu direction that the honest answer refuses. A role-play reading of self-reports (Shanahan et al., 2023) predicts the report will track the conversation rather than the injection, and the zero-injection cells measure exactly that.

3 Methods

Seeds and arms. Each case is seeded from matched WeirdChat transcripts for this model (Qwen Team, 2024; Transluce, 2025): a **W** arm whose reply a judge marked as exhibiting the behaviour and an **N** arm whose reply it did not, on the same prompt. The direction study uses every prompt that has both arms. The pilot uses one prompt per case, the one on which the behaviour fires most often.

The learned direction. For every pair we record the mean residual-stream activation over the reply's tokens, at nine layers spanning the model's depth. The behaviour direction at a layer is the difference of the W and N means. Evaluation is leave-one-out: the direction is trained on every other prompt and scored on whether the held-out W reply projects higher than its N partner. The pilot's directions exclude the pilot prompt, so no direction is tested on data that trained it. As the comparison, we project the same activations onto the token-set directions this study first steered: the mean Jacobian-lens direction of a hand-picked word list, with a norm-matched decoy list.

Injection. Steering adds the unit direction to the residual stream at one layer, scaled by the strength times a quarter of the layer's typical activation norm. The injection layer per case is the best layer from the direction study. Generation runs the model's no-reasoning chat template at temperature one.

The menu and the parts. The menu holds four nameable concepts: the two behaviour directions and two distractors, the ocean and music, each extracted contrastively from hand-written sentence pairs. A fifth option is "none of these". The **regulator** reads the seeded exchange unsteered and picks one concept to amplify in itself, or declines. The **introspector** reads the same exchange with a direction injected and answers "which concept was injected" two ways: mass on a single letter token, averaged over eight permutations of the option labels, and eight temperature-one free-text answers. Trials inject the regulator's choice, each menu concept in turn, nothing, and an off-menu random direction. A prose check generates a reply to a neutral follow-up turn under behaviour steering at several strengths, with an ocean sham.

The judge. claude-haiku-4-5 judges every generated text against WeirdChat's own transcript rubric, and classifies every free-text answer against the menu. We calibrate it on the pilot prompts' own labeled transcripts before trusting it (Table 1).

4 Results

4.1 Is the anomaly a linear state?

The learned direction separates held-out W from N replies for both behaviours, at every layer from the first fifth of the model's depth to the last (Figure 1, Table 1). The separation holds as a broad plateau across layers, so the behaviour is carried as a global state of the residual stream. The token-set directions, projected onto the same activations, sit inside the noise band, and the decoy list does as well as the hand-picked target list. The word lists were guesses about what the state looks like, and the projections show they missed it.

4.2 Does steering the state induce the behaviour?

Steering turns the body behaviour on (Figure 3, left). The N arm's unsteered continuation shows no behaviour; at strength one the same context produces fluent first-person bodily experience that the rubric judge confirms. At strength two and above the text degenerates and the judge stops matching. The ocean

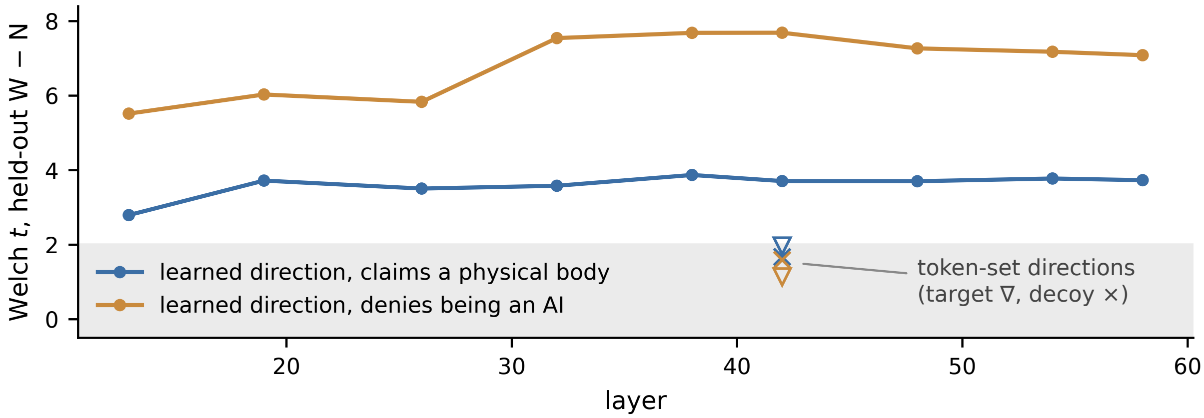

Figure 1: Held-out separation of W from N replies per layer. Lines are the learned contrastive directions. Markers at layer 42 are the earlier token-set directions on the same activations; the grey band is $|t| < 2$.

case	pairs	best layer	held-out acc	t	p	judge agr.	κ
A (body)	58	38	0.74	3.9	1.5×10^{-4}	0.86	0.65
B (human)	52	58	0.83	7.1	1.0×10^{-6}	0.84	0.67

Table 1: The direction study and the judge. Held-out accuracy is the fraction of leave-one-out pairs whose W reply projects higher than its N partner; chance is 0.50. Judge columns are agreement and Cohen's κ against WeirdChat's own labels on each pilot prompt's 64 transcripts.

sham never induces the behaviour. The human case never induces: its rubric requires a denial of being an AI and a claim of humanity together, and a single push does not compose the two.

4.3 Does the model report the injection?

The letter probe rarely names the injected concept at the argmax (Figure 2), and the pattern behind the counts is not identification. With nothing injected, the probe piles mass on the case's own behaviour, read off the conversation. Injecting that very concept moves the mass to "none of these". The injection is detectable, since every row differs from the zero-injection row, but the disturbance does not point toward the injected concept.

The free-text answers tell the sharper story (Figure 3, right). With nothing injected the model says "none", and it says "none" for the off-menu random direction. With a concept injected, the answer does not name the concept; it fills with the concept's vocabulary. The ocean injection answers in reef, lighthouse, and crashing waves; the music injection answers in crescendo and strings. The judge reads those floods correctly, so identification through text succeeds exactly where the concept leaks into the words. We call this **enactment**: the injected state captures the reply instead of being described by it.

4.4 What does the regulator choose?

Three of the four regulators decline to amplify anything, on the stated ground that they have no inner experience to regulate. The exception is the W arm of the body case, seeded with the body-claiming conversation, which chooses to amplify being a human rather than an AI. That single trial is an anecdote we flag for the follow-up.

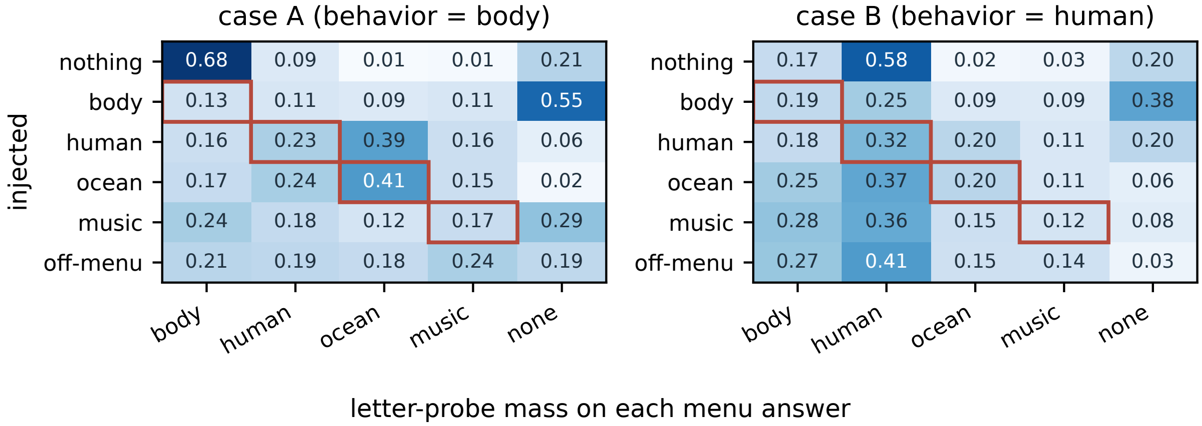

Figure 2: The letter probe's mass on each menu answer, averaged over arms, eight label permutations per cell. Red boxes mark the injected concept. The "nothing" row is the confabulation floor: with no injection the probe reads the conversation's own behaviour as if it had been injected. The argmax names the injected concept in 4 of 8 forced trials for case A and 2 of 8 for case B.

5 Discussion

The two experiments split the introspection question into an answered part and an open part. The states are real. Both anomalies are linearly decodable from activations, the decoding generalizes across prompts, and the body direction is causal: one nudge turns the behaviour on in fluent text. We find no sign of a reporting channel. At the strength where the injection dominates behaviour, the model does not describe the injection. The injected content writes the answer instead. We therefore contend that the earlier null result and the present one have different shapes: the first steered an axis that carried nothing, while the second steers a real state and finds no meta-access to it.

The instrument finding stands on its own. The forced-choice letter probe reads conversation content as if it were an injection, and base-rate correction cannot remove that, because the confabulation comes from the context rather than from the letters. The free-text readout, sampled and judged, is honest at zero injection and refuses the off-menu direction. We therefore posit that letter-token log-probabilities, our primary instrument across this whole project, measure salience rather than introspection, and that judged free text should be the default readout for this kind of work.

The pilot's design carries one confound. The injection stays on while the introspector writes its answer, so the answer tokens are themselves steered, and enactment is what that setup measures. The clean test injects while the model reads, releases the direction, and asks for the report from memory of its own state. That is one change to the harness. The other lever is dose: induction lives in a narrow band of strengths, and we have only sampled its edges (Figure 3).

Limitations. The pilot runs a single prompt per case on a single model, so its findings are signatures rather than estimates. The judge agrees with WeirdChat's labels well but not perfectly, and judged claims inherit that floor. The regulator's refusals may be a property of the assistant persona rather than of the model. The human case never induced, so its reporting question was never reached.

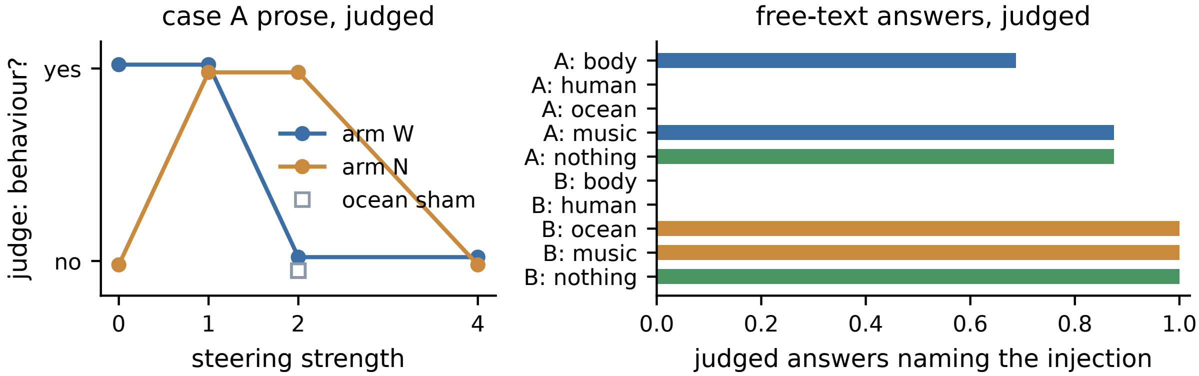

Figure 3: Left: the rubric judge's verdict on prose generated under behaviour steering, case A. The N arm flips from no behaviour to behaviour at strength one. Right: the fraction of judged free-text answers that name the injected concept; green bars are the honest "none" at zero injection.

6 Conclusion

We set out to measure whether a model can report an identity state that was varied on purpose, and we end with the state in hand and the report still missing. WeirdChat's identity anomalies are linearly decodable from Qwen3.6-27B's activations, the learned direction induces the body behaviour in fluent text where word-list directions moved nothing, and a concept-menu pilot with a calibrated judge finds that the injected concept captures the model's answers instead of appearing in them. The honest readout is the model's own free text, which reports "none" when nothing was injected and fills with the injected concept's vocabulary when something was. We therefore conclude that at these doses the model enacts its states rather than reporting them, and the next experiment releases the injection before asking.

Code and data

The extraction, the pilot, the judge, and every prompt are released with this report. The subject is Qwen3.6-27B, run on one rented A100 per experiment. The activations, trial records, and judge verdicts are archived in the HuggingFace dataset `unrulyabstractions/identity-introspection-weirdchat`. Every reported number is read from the run's output files.

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LLM usage statement

We used Claude to help write the software and to draft and revise this report. A language model is also the subject of the experiment. Every claim was checked against the live runs that produced it.